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Roll no. = 02

Subject = Full Stack lab

Lab Sheet 4

#TASK 1

SUM.js

```
function sum(a, b) {
    return a + b;
} module.exports = sum;
```

Even-odd.js

```
function evenOdd(number) {
    return number % 2 === 0 ? "Even" :
"Odd"; }

module.exports = evenOdd;
```

App1.js

```
const sum = require("./sum");
const evenOdd = require("./even-odd");
console.log("Sum:", sum(10, 20));
console.log("15 is", evenOdd(15));
console.log("24 is", evenOdd(24));
```

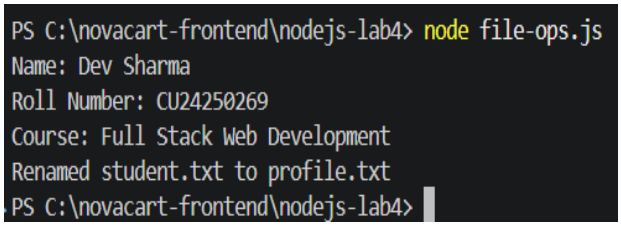
```
PS C:\novacart-frontend\nodejs-lab4> node app1.js
Sum: 30
15 is Odd
24 is Even
PS C:\novacart-frontend\nodejs-lab4>
```

#Task 2

file-ops.js

```
const fs = require("fs");
const fileName = "student.txt";
const newFileName = "profile.txt";
fs.writeFile(fileName, "Name: Dev Sharma\nRoll Number: CU24250269\n", (err) => {
    if (err) throw err;
  fs.appendFile(fileName, "Course: Full Stack Web Development\n", (err) => {  if (err) throw err;
```

```
  fs.readFile(fileName, "utf8", (err, data) => {   if (err) throw err;
console.log(data.trim());  fs.rename(fileName, newFileName, (err) => {
 if (err) throw err;  console.log(`Renamed ${fileName} to ${newFileName}`);
          });
       });
   });
});
```



#Task 3

dir-ops.js

```
const fs = require("fs");
const folder = "uploads";
const files = ["file1.txt", "file2.txt", "file3.txt"];
function createFiles() {  let remaining = files.length;   files.forEach((file) => {
  fs.writeFile(`${folder}/${file}`, "", (err) => { if (err) throw err;
    remaining--;
if (remaining === 0) {
fs.readdir(folder, (err, entries) => { if (err) throw err;
console.log("Files:", entries.join(", ")); fs.unlink(`${folder}/${files[0]}`, (err) => {
if (err) throw err;  console.log(`Deleted: ${files[0]}`);
}); }); }
        });
    });
}
fs.mkdir(folder, { recursive: true }, (err) => {
    if (err) throw err;
    createFiles(); });
```

```
PS C:\novacart-frontend\nodejs-lab4> node dir-ops.js
Files: file1.txt, file2.txt, file3.txt
Deleted: file1.txt
PS C:\novacart-frontend\nodejs-lab4>
```

#Task 4

path-demo.js

```
const path = require("path");

const filePath = "/home/user/data/report.pdf";

const relativePath = "./data/report.pdf";

console.log("Directory:", path.dirname(filePath));

console.log("Base name:", path.basename(filePath));

console.log("Extension:", path.extname(filePath));

console.log("Absolute path:", path.resolve(relativePath));
```

```
PS C:\novacart-frontend\nodejs-lab4> node path-demo.js
Directory: /home/user/data
Base name: report.pdf
Extension: .pdf
Absolute path: C:\novacart-frontend\nodejs-lab4\data\report.pdf
PS C:\novacart-frontend\nodejs-lab4>
```

#Task 5

sys-report.js

```
const os = require("os");

console.log("Platform:", os.platform());

console.log("Architecture:", os.arch());

console.log("Total Memory:", `${(os.totalmem() / 1024 ** 3).toFixed(2)} GB`);

console.log("Free Memory:", `${(os.freemem() / 1024 ** 3).toFixed(2)} GB`);

console.log("CPU Cores:", os.cpus().length);

console.log("Uptime:", `${(os.uptime() / 60).toFixed(2)} minutes`);
```

```
PS C:\novacart-frontend\nodejs-lab4> node sys-report.js
Platform: win32
Architecture: x64
Total Memory: 15.70 GB
Free Memory: 3.99 GB
CPU Cores: 12
Uptime: 1554.63 minutes
PS C:\novacart-frontend\nodejs-lab4>
```

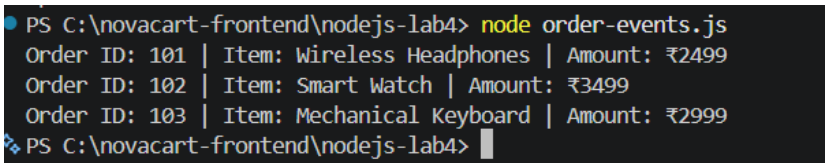
#Task 6

order-events.js

```
const EventEmitter = require("events").EventEmitter;

const orderEvents = new EventEmitter();
```

```
orderEvents.on("orderPlaced", ({ id, item, amount }) => {
    console.log(`Order ID: ${id} | Item: ${item} | Amount: ₹${amount}`); });
orderEvents.emit("orderPlaced", { id: 101, item: "Wireless Headphones", amount: 2499 });
orderEvents.emit("orderPlaced", { id: 102, item: "Smart Watch", amount: 3499 });
orderEvents.emit("orderPlaced", { id: 103, item: "Mechanical Keyboard", amount: 2999 });
```



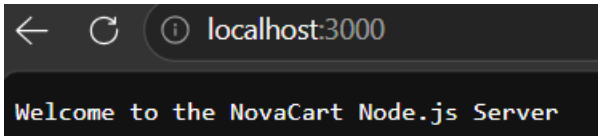
#Task 7

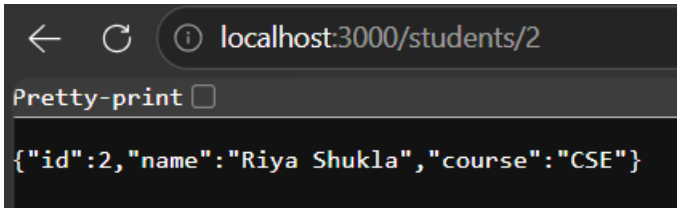
node server.js

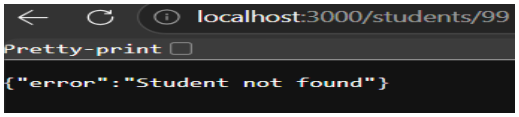
```
const http = require("http");
const url = require("url");
const students = [
    { id: 1, name: "Dev Sharma", course: "CSE" },
    { id: 2, name: "Riya Shukla", course: "CSE" },
    { id: 3, name: "Aabra ka dabra", course: "CSE" }
];
const server = http.createServer((req, res) => {
    const parsedUrl = url.parse(req.url, true);
    const pathname = parsedUrl.pathname;
   res.setHeader("Content-Type", "application/json");
 if (req.method !== "GET") {
        res.statusCode = 405;
        res.end(JSON.stringify({ error: "Method Not Allowed" }));
        return;
    }
  if (pathname === "/") {
        res.setHeader("Content-Type", "text/plain");
        res.end("Welcome to the NovaCart Node.js Server");
        return; }
  if (pathname === "/students") {  res.end(JSON.stringify(students));
        return;
```

```
    }
const studentMatch = pathname.match(/^\/students\/([^/]+)$/);
    if (studentMatch) {
        const id = Number(studentMatch[1]);
        const student = students.find((item) => item.id === id);
if (!student) {
            res.statusCode = 404;
            res.end(JSON.stringify({ error: "Student not found" }));
            return:  }
res.end(JSON.stringify(student));
        return:  }
if (pathname === "/search") {
        const keyword = parsedUrl.query.keyword || "";
        res.setHeader("Content-Type", "text/plain");
        res.end(`Search keyword: ${keyword}`);
        return; }
res.statusCode = 404;
    res.setHeader("Content-Type", "text/html");
    res.end("<h1>404 - Page Not Found</h1>");
});]
const PORT = 3000;
server.listen(PORT, () => {
    console.log(`Server running at http://localhost:${PORT}`);
});
```

```
PS C:\novacart-frontend\nodejs-lab4> node server.js
Server running at http://localhost:3000
```

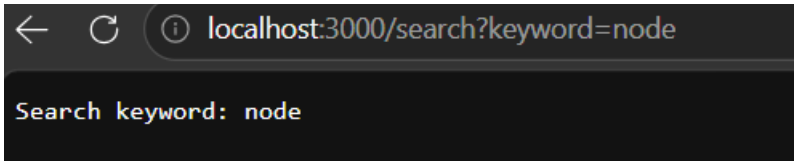






#Task 8

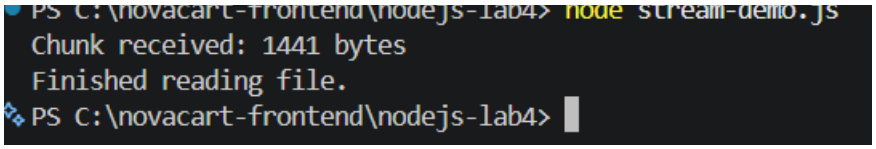
node server.js



#Task 9

stream-demo.js

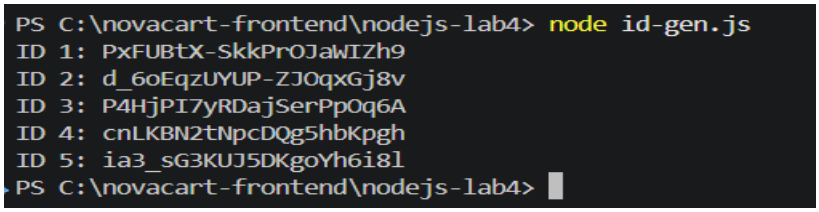
```
const fs = require("fs");
const fileName = "large-file.txt";
let content = "";
for (let i = 1; i <= 50; i++) {
    content += `This is line ${i} of the file.\n`; }
fs.writeFile(fileName, content, (err) => {
    if (err) throw err;
  const stream = fs.createReadStream(fileName);
  stream.on("data", (chunk) => {
        console.log(`Chunk received: ${chunk.length} bytes`);
    });
stream.on("end", () => {
        console.log("Finished reading file.");
    }); });
```



#Task 10

id-gen.js

```
const { nanoid } = require("nanoid");
for (let i = 1; i <= 5; i++) {
    console.log(`ID ${i}: ${nanoid()}`);
}
```



#Task 11

calc-cli.js

```
const [, , first, second, operator] = process.argv;
const a = Number(first);
const b = Number(second);
let result;
if (Number.isNaN(a) || Number.isNaN(b)) {
    console.log("Please enter two valid numbers.");
    process.exit(1);
}
switch (operator) {
    case "+":
        result = a + b;
        break;
    case "-":
        result = a - b;
        break;
    case "*":
        result = a * b;
        break;
    case "/":
        if (b === 0) {
            console.log("Cannot divide by zero.");
            process.exit(1);  }
        result = a / b;
        break;
    default:
        console.log("Use +, -, *, or / as the operator.");
        process.exit(1);
```

```
}
console.log(`Result: ${result}`);
```

```
PS C:\novacart-frontend\nodejs-lab4> node calc-cli.js 10 5 +
Result: 15
PS C:\novacart-frontend\nodejs-lab4>
```

```
PS C:\novacart-frontend\nodejs-lab4> node calc-cli.js 10 5 -
Result: 5
PS C:\novacart-frontend\nodejs-lab4>
```

```
PS C:\novacart-frontend\nodejs-lab4> node calc-cli.js 10 5 "*"
Result: 50
PS C:\novacart-frontend\nodejs-lab4>
```

```
PS C:\novacart-frontend\nodejs-lab4> node calc-cli.js 10 5 /
Result: 2
PS C:\novacart-frontend\nodejs-lab4>
```

#Task 12

env-demo.js

```
const http = require("http");
const PORT = Number(process.env.PORT) || 4000;
const server = http.createServer((req, res) => {
    res.writeHead(200, { "Content-Type": "text/plain" });
    res.end(`Server is running on port ${PORT}`);
});
server.listen(PORT, () => {
    console.log(`Active port: ${PORT}`);
});
```

```
PS C:\novacart-frontend\nodejs-lab4> $env:PORT=5000
>> node env-demo.js
Active port: 5000
```

